

Beam B12 - Level 3 slab

Project & Basis	
Calculator	RCC Beam - IS 456:2000
Project	-
Country / Code	India IS 456:2000
Result	-
Design Inputs	
span m	6
Mu kNm	180
Vu kN	140
b mm	300
fck	30
fy	500
Section	
b mm	300
D mm	350
d mm	309
cover mm	25
Moment Capacity	
Mu lim kNm	114.05
applied Mu kNm	180
is singly	No
Tension Steel	
Ast required mm2	1630.8
Ast min mm2	157.6
bars	9 - #16
Ast provided mm2	1809.9
pt percent	1.952
Compression Steel	
required	Yes
Asc mm2	749.8
bars	4 - #16
Shear Design	
?v N mm2	1.51
?c N mm2	0.836
?c max N mm2	3.5
mode	Stirrups required
stirrups	2L - #8 @ 216 c/c

stirrup spacing mm

216

warning

Deflection

basic span d ratio

20

d provided mm

309

d required mm

300

ok

Yes

Development Length

Ld mm

725

bar dia mm

16

Summary

design type

Doubly Reinforced

beam type

simply_supported

code

IS 456:2000

Proof & Official Sources

Code Basis

IS 456:2000

Maturity

Preliminary design calculator

Clauses / tables: IS 456 flexure; IS 456 shear; IS 456 deflection; IS 456 detailing

Bureau of Indian Standards (BIS) - Government of India / National Standards Body

<https://www.bis.gov.in/>

Assumptions: Preliminary RCC beam design. Verify ductile detailing, development length, torsion, serviceability, exposure, and final bar arrangement.

Formula trace and official source link included; requires project review by a licensed engineer before construction.

Building law is adopted by the project jurisdiction. Verify local amendments, national annexes, authority requirements, and the enforced edition.

Certification & Sign-off

Prepared

Demo User

06 Jul 2026 01:21

Checked

R. Krishnan (Checker)

06 Jul 2026 01:21

Approved

S. Mehta, P.E. (Approver)

06 Jul 2026 01:21

This is a preliminary engineering calculation. Figures must be independently verified and the sheet sealed by the responsible licensed engineer under the governing jurisdiction before use in construction.

